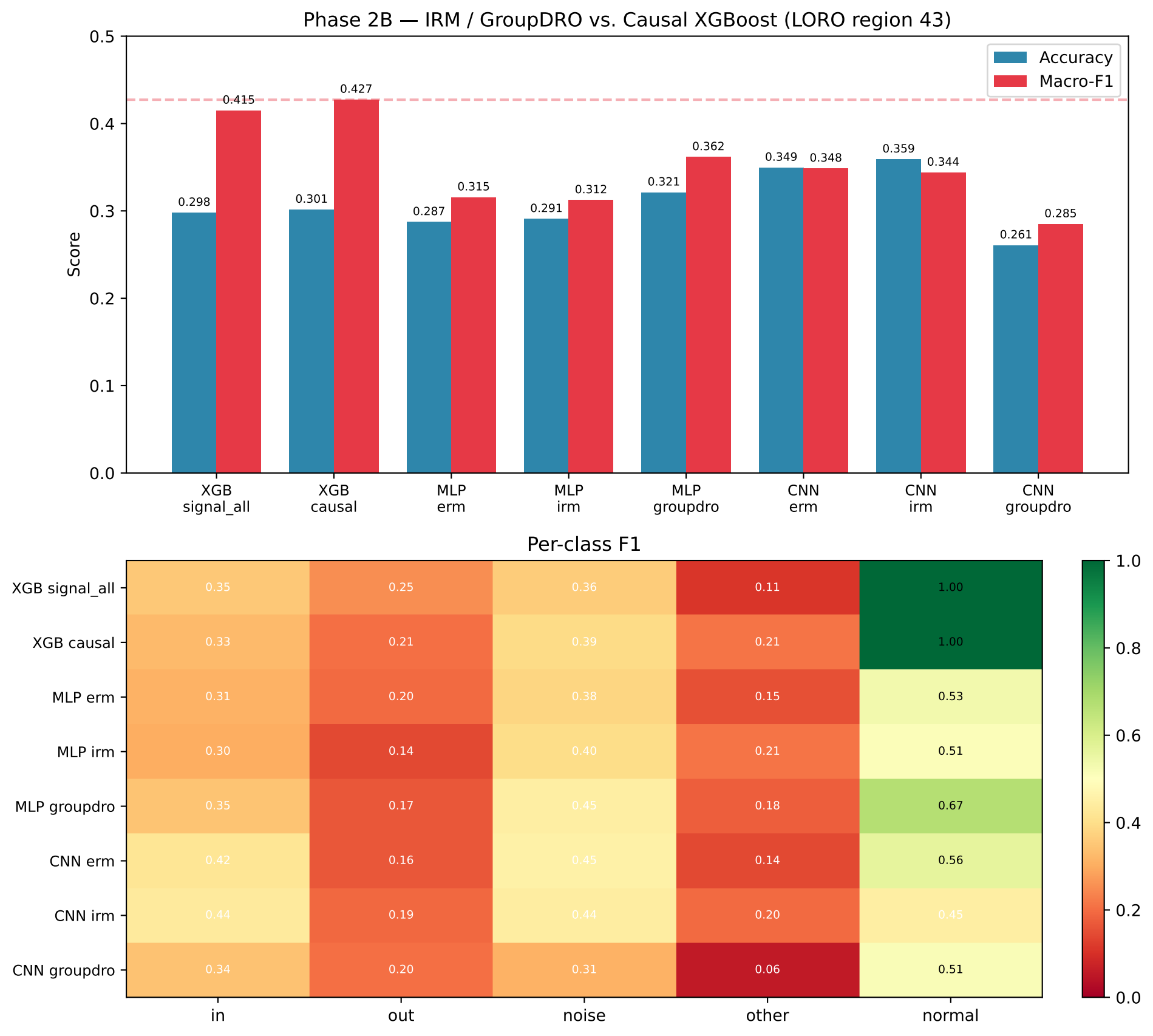
Phase 2B — IRM / GroupDRO vs. Causal XGBoost (LORO region 43)
Accuracy
Macro-F1
Score
0.0
0.1
0.2
0.3
0.4
0.5
0.298
0.415
0.301
0.427
0.287
0.315
0.291
0.312
0.321
0.362
0.349
0.348
0.359
0.344
0.261
0.285
XGB signal_all
XGB causal
MLP erm
MLP irm
MLP groupdro
CNN erm
CNN irm
CNN groupdro
Per-class F1
XGB signal_all
XGB causal
MLP erm
MLP irm
MLP groupdro
CNN erm
CNN irm
CNN groupdro
0.35 0.25 0.36 0.11 1.00
0.33 0.21 0.39 0.21 1.00
0.31 0.20 0.38 0.15 0.53
0.30 0.14 0.40 0.21 0.51
0.35 0.17 0.45 0.18 0.67
0.42 0.16 0.45 0.14 0.56
0.44 0.19 0.44 0.20 0.45
0.34 0.20 0.31 0.06 0.51
in
out
noise
other
normal
1.0
0.8
0.6
0.4
0.2
0.0